**The Assessment Question Paper and Assessment Answer Booklet must not be removed from the assessment room. Both must be handed in at the end of the assessment.**

SIM Global Education

# WRITTEN TEST
# CONTINOUS ASSESSMENT (MAIN)

## (DIPLOMA IN INFORMATION TECHNOLOGY)

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|---|---|---|
| **Date** | **:** | **23 April 2025** |
| **Module** | **:** | **ITS D002 Problem Solving** |
| **Time** | **:** | **10:00 – 11:30** |
| **Duration** | **:** | **1 hour 30 minutes** |
| **Total no. of pages (including this cover page)** | **:** | **5** |

**INSTRUCTIONS TO CANIDATES:**

1. This assessment carries a total of 100 marks. Answer **ALL** questions:

   **Section A (50 marks): 2 Structural Questions**
   **Section B (50 marks): 2 Application Questions**

2. Answer **ALL** questions in the answer booklet provided. Fill in the cover page carefully. **You will receive a zero (0) score if your name is filled incorrectly**.

3. All answers must be clearly written. Begin each question on a fresh page.

4. The use of non-programmable calculators is **NOT** permitted in this assessment.

5. This is a closed-book assessment. No unauthorised materials are allowed in the assessment venue.

**At the end of the assessment:**
Please ensure that you have written your name on each answer booklet used: failure to do so will mean that your work cannot be identified. If you have used more than one answer book, please tie them together with the string provided.

**SIM GLOBAL EDUCATION RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.**

SAMPLE

## SECTION A

### Question 1

Botanics Singapore, a company famous for selling beautiful orchid flowers, recently moved to a new location where they could grow and sell more orchids. The new location allowed Botanics Singapore to grow more than 5000 orchids. However, soon after moving, a lof of the orchids started to wilt and die. The company is desperate to find a solution, in order to save the rest of the orchids and reduce losses. Orchids can grow well in Singapore, but are sensitive to the following conditions:

1) Ideal temperature for orchids is 20 to 24 ºC. Should not be exposed to too much sunlight.
2) Too much water. Water collecting at the roots will damage the plant.
3) Attack by fungus.
4) Damage during movement.

You have been assigned to solve the problem of the dying orchids.

**Required:**

(a) The first step to Problem Solving is first representing the solution. What are the states that have to be identified? Answer with respect to the scenario above.

(5 marks)

(b) Describe the Divide and Conquer problem solving method by stating its advantages and disadvantages. How can it help to solve the problem?

(10 marks)

(c) Describe **TWO (2)** other problem solving methods and how they can help to solve the problem.

(10 marks)

SAMPLE

**Question 2**

Given a list of data: 12, 17, 4, 32, 22, 15 stored in an array (or list), answer (a) and (b). State clearly the values of the indices/positions and the values of the array during the process of sorting with relevant pictures or diagrams.
(Note: you do not need to write the algorithm/code)

**Required:**

(a) Describe Linear Search for the value '18' from start to finish.

(10 marks)

(b) Would you be able to use Binary Search on the above data? Explain.

(5 marks)

(c) Describe Merge Sort from start to finish.

(10 marks)

SAMPLE

## SECTION B

### Question 3

Botanics Singapore wants to count the number of orchids that need urgent treatment. Every orchid is given a rating from 1 to 5 (1 being very healthy, 5 is dying). Orchids that are rated 3 or 4 can be given treatment, but those rated 5 can be thrown away. Every orchid is given a unique number to identify them.

**Required:**

(a) Draw a flowchart for entering the orchid number, colour of the orchid and the orchid rating for ONE orchid only. The flowchart should output one of these 3 messages based on the rating:

- "Orchid [XXX] is healthy"
- "Orchid [XXX] needs treatment"
- "Orchid [XXX] can be thrown away"

where [XXX] is the orchid number entered by the user.

(10 marks)

(b) Write in Pseudocode a program that will allow the user to enter the same information as part (a), but for 10 orchids. The program should also output the number of orchids that need treatment. (You are required to use Loops)

(15 marks)

SAMPLE

## Question 4

Botanics Singapore wants to create a program to record the status of their orchids, and a report of how many orchids need treatment. The output at the end of the program will show the orchid numbers those that need treatment, the numbers of those to be thrown away, and the status with the highest number of orchids. Refer to Question 3 on the rating system on orchids.

Below is a sample run of the program.

[*user inputs are in* ***BOLD***]

--- Orchid Record system ----
Enter orchid number: **156**
Enter status: **1**
Add another record? **y**

Enter orchid number: **157**
Enter status: **3**
Add another record? **y**

Enter orchid number: **158**
Enter status: **4**
Add another record? **y**

Enter orchid number: **159**
Enter status: **1**
Add another record? **y**

Enter orchid number: **160**
Enter status: **5**
Add another record? **y**

Enter orchid number: **161**
Enter status: **1**
Add another record? **n**

Needs treatment: 157, 158
Throw away: 160
Status with the most orchids: 1

**Required:**
Code the program in Python. Do indicate useful comments in your codes, and take note that indentation is crucial.

(25 marks)

**---END OF PAPER---**

SAMPLE